M&AE 4070/5070 - Dynamics of Flight Vehicles

Case Study 4 - SOLUTIONS

Take-off and Landing

In this case study, you will go through the various phases of take-off and landing of a commercial transport aircraft, with the specific objective to determine the runway length necessary for safe take-off and landing. You will evaluate how the minimum runway length depends on the aircraft aerodynamic configuration, in particular with respect to high-lift systems. Engine failure being the most detrimental event an aircraft can experience at take-off, we will take this configuration as reference case.

Available data The aircraft considered in this case study is modeled after a typical older generation, long-haul aircraft.

- **Drag polar**: $C_x = C_{x_0} + kC_z^2$, with $k = 0.055$ and $C_{x_0} = 0.0175$.
- **Engines and thrust**: the aircraft has $n = 2$ engines, and for the purpose of this case study, we consider a slightly more detailed model for the thrust. Assume that the engines provide a combined total thrust that depends on the Mach number according to:

$$F(M) = n\frac{\rho}{\rho_0}F_0\left(1 - M + \frac{M^2}{2}\right)\delta_x \tag{1}$$

with

$$F_0 = 1.85 \times 10^5 \text{ N} \tag{2}$$

- **Mass:** $m = 1.2 \times 10^5$ kg $= 120$ tons.
- **Reference surface area:** $S = 260\ m^2$.
- **Lift coefficient** C_z:

$$C_z = C_{z_\alpha}\bar{\alpha} + C_{z_{\delta_m}}\delta_m + \Delta C_{z,SP} \tag{3}$$

The C_z gradient C_{z_α} is equal to 5. In $C_{z_\alpha}\bar{\alpha}$, we find the effects of the slats and flaps through the modification of the angle of attack at zero lift, α_0 (comes from the flaps), and the maximum angle of attack, $\alpha_{\max}$ (comes from the slats). $\bar{\alpha}$ therefore is written

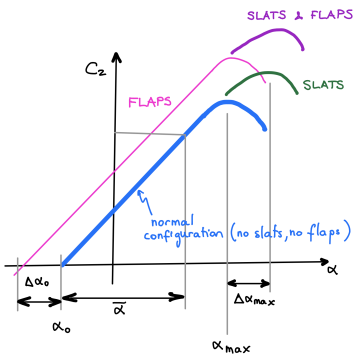
$$\bar{\alpha} = \alpha - \alpha_0 \tag{4}$$

where

$$\alpha_0 = \alpha_{0,\text{ no SF}} + \Delta\alpha_{0,\text{flaps}} \tag{5}$$

$$\alpha_{\max} = \alpha_{\max,\text{ no SF}} + \Delta\alpha_{\max,\text{ slats}} \tag{6}$$

Those contributions are sketched in the following figure:



In this specific case study, we assume that the moment equation is satisfied and that the $C_{z_{\delta_m}}\delta_m$ contribution to C_z is small enough that it can be neglected.

- **Additional data needed to calculate lift and drag coefficients for various aircraft configurations**:

Configuration	#	$\alpha_{\max}$	α_0	$\bar{\alpha}_{\max}$
Nothing deployed	0	14°	-2°	16°
Slats only, 18°	1	18°	-2°	20°
Slats 18°, Flaps 10°	2	18°	-8°	25.7°
Slats 22°, Flaps 15°	3	19.3°	-11.4°	30.7°
Slats 22°, Flaps 20°	4	19.3°	-16.6°	35.9°

Table 1: Angles of attack modification from slats and flaps deployment.

Configuration	ΔC_{x_0}	$\Delta C_{z,SP}$
Landing gear	0.0175	0
Spoilers - landing	0.06	-0.15

Table 2: Additional drag and lift coefficients from landing gears and spoilers

Configuration	#	ΔC_{x_0}
Nothing deployed	0	0
Slats only, 18°	1	0.0118
Slats 18°, Flaps 10°	2	0.0185
Slats 22°, Flaps 15°	3	0.0243
Slats 22°, Flaps 20°	4	0.0583

Table 3: Additional drag coefficients from high-lift systems

1 Take-off

The aircraft is positioned at the beginning of the runway, ready to take off. The pilot goes full throttle while keeping the brakes engaged. They release the brakes.

1. Starting at that time, calculate the runway length needed for the acceleration, the rotation and positioning onto a linear climb trajectory, and the linear climb to safety height h_s for all configurations listed in the tables above, from #1 (nothing deployed) to #4 (slats and flaps fully deployed, that is, find:

$$L_r = L_{r_{\text{acc}}} + L_{1/2\text{ flare}} + L_{r_{\text{climb}}} \tag{7}$$

   A Python/Matlab script and a table form to provide the results are strongly suggested.

2. Comment on the results, in particular, compare them to the one provided by the *magic line* formula given in lecture notes.

3. You should observe that take-off with some configurations is just not possible: why? Show that a higher rotation velocity V_R would help solve the problem, even though it would increase the distance spent on the ground.

Assume for simplicity that:

- The runway is horizontal ($\gamma_r = 0$)
- The taxiing angle of attack $\alpha_{a,r} = 0$.
- The friction coefficient σ_r is constant and equal to $1.5 \times 0.015 = 0.0225$ (dry concrete)
- The velocity at rotation $V_R = 1.15V_S$.
- The acceleration between the start of the rotation and the time at which V_{LOF} is negligible, leading to $V_R \approx V_{LOF}$

As the runway length is typically not a problem unless there is an engine failure, we'll consider the critical case of engine failure occurring when the aircraft has a velocity equal to V_1. V_1 is typically obtained via an iterative procedure, so instead, we'll fix V_1 at a typical value for twin-engines: $V_1 = 0.933V_R$.

- Use an average acceleration $\Gamma_{0.7}$, evaluated at a velocity $0.7V_R$ to evaluate the distance $L_{r_{\text{acc,V}_1}}$ needed to reach V_1.
- At V_1, one engine fails. Calculate the distance $\Delta L_{r_{\text{acc}}}$ the aircraft covers before reaching $V_{\text{LOF}} = V_R$, taking the acceleration equal to the one evaluated at V_1 with one engine out.
- Add the two contribution to obtain $L_{r_{\text{acc}}}$:

$$L_{r_{\text{acc}}} = L_{r_{\text{acc,V}_1}} + \Delta L_{r_{\text{acc}}} \tag{8}$$

Solutions

The total runway length for take-off (with engine failure at V_1) is:

$$L_r = L_{r_{\text{acc}}} + L_{1/2\text{ flare}} + L_{r_{\text{climb}}} \tag{9}$$

where $L_{r_{\text{acc}}} = L_{r_{\text{acc,V}_1}} + \Delta L_{r_{\text{acc}}}$.

Acceleration phase. The longitudinal acceleration on the runway is (with $\alpha_{a,r} = 0$, $\gamma_r = 0$):

$$\Gamma = \frac{F}{m} - \frac{\frac{1}{2}\rho S V^2}{m}\left(C_x - \sigma_r C_z\right) - g\sigma_r \tag{10}$$

with $\sigma_r = 0.0225$, $C_x = C_{x_0} + kC_z^2$, $k = 0.055$ (given). During acceleration with $\alpha = 0$: $\bar{\alpha} = -\alpha_0$, hence $C_z = C_{z_\alpha}(-\alpha_0)$. The distances are:

$$L_{r_{\text{acc,V}_1}} = \frac{V_1^2}{2\Gamma_{0.7V_R}} \quad \text{(2 engines, evaluated at } 0.7V_R\text{)} \tag{11}$$

$$\Delta L_{r_{\text{acc}}} = \frac{V_R^2 - V_1^2}{2\Gamma_{V_1}} \quad \text{(1 engine, evaluated at } V_1\text{)} \tag{12}$$

Climb phase. From the force balance: $\dfrac{F}{mg} = \dfrac{1}{f} + \gamma$, hence

$$\gamma = \frac{F}{mg} - \frac{1}{f} \tag{13}$$

At lift-off, $C_z = (V_R/V_S)^2 C_{z,\max} = C_{z,\max}/1.15^2$. The lift-to-drag ratio is $f = C_z/C_x$.

$$L_{r_{\text{climb}}} = \frac{h_s}{\gamma} \tag{14}$$

with $h_s = 10.7$ m (safety height).

Flare. The equation for the (half) flare is

$$L_{1/2\text{ flare}} = \frac{1}{n_f - 1} \frac{V_{\text{LOF}}^2}{2g} \gamma_{\text{TO}} \tag{15}$$

with $n_f = 1.1$ and γ_{TO} from the climb analysis.

Aerodynamic coefficients. The stall velocity is $V_S = \sqrt{2mg/(\rho S C_{z,\max})}$ with $C_{z,\max} = C_{z_\alpha} \bar{\alpha}_{\max}$. For each configuration, C_{x_0} includes: base drag (clean aircraft ≈ 0.0175), landing gear ($\Delta C_{x_0} = 0.0175$), and high-lift system contribution from the table. The reference surface is $S = 260$ m^2 and $C_{z_\alpha} = 5$.

Results. Using a python script to automate calculations, we obtain the following values for each configuration.

Config	V_S (m/s)	V_R (m/s)	$F/(mg)$ (1 eng)	γ (deg)	$L_{\text{acc,V}_1}$ (m)	ΔL_{acc} (m)	L_{flare} (m)	L_{climb} (m)	L_r (m)
0	72.8	83.7	0.123	1.83	1368	610	114	334	2427
1	65.1	74.8	0.126	1.05	1078	485	52	584	2199
2	57.4	66.0	0.130	0.27	829	381	11	2239	3459
3	52.5	60.4	–	–	695	337	impossible	impossible	impossible
4	48.6	55.9	–	–	626	432	impossible	impossible	impossible

where $F/(mg)$ is the thrust-to-weight ratio of the single engine at lift-off (V_R), and γ is the climb slope (degrees).

Comments.

1. **Comparison with magic line:** The magic line formula is an empirical relation roughly of the form: $L_r \approx k V_S^2$. Using the oneprovided in the lecture note, we find that it gives reasonable runway lengths (within a few percents) for configurations with limited high-lift deployment. Predictions for larger flaps and slats angles are very poor.
2. **Impossible configurations:** Configurations 3 and 4 are *impossible* at $V_R = 1.15 V_S$: $\gamma = F/(mg) - 1/f \leq 0$. The single-engine thrust cannot overcome drag at the required lift coefficient. The lift-to-drag ratio f drops significantly with heavy high-lift deployment.
3. **Higher V_R:** Increasing V_R reduces C_z at lift-off, hence increases f and can make $\gamma > 0$. For config 3: at $V_R = 1.3 V_S$, take-off becomes possible ($L_r \approx 4043$ m, $\gamma \approx 0.24°$); at $V_R = 1.4 V_S$, $L_r \approx 3176$ m, $\gamma \approx 0.47°$. The trade-off is longer ground roll, but climb becomes feasible.

To optimize take-off, both deployment of high-lift systems and rotation/LOF velocity must be optimized!

2 Landing

Let's evaluate now the runway length needed when landing. We assume that the aircraft lands with the same mass as for take-off. After all, we assumed engine failure, so it makes sense! The aircraft configuration is #4, with high-lift systems fully deployed. Landing comprises three phases:

- an aerial phase, during which the aircraft needs to pass the safety height of 15.2 m with a standard slope of 3°(ILS).
- the landing flare (in fact, half a flare), covering a distance of:

$$L_{1/2\text{ flare}} = \frac{1}{n_f - 1} \frac{V_{\text{app}}^2}{2g} \gamma \tag{16}$$

  where the approach velocity V_{app} is taken equal to 1.3 V_S, and the load factor n_f is 1.1. During this phase, the velocity decreases until it reaches its touch down value $V_{TD} = 1.15 V_S$.

- Finally, the last phase is the braking phase. We assume that thrust is zero, with the same runway conditions than during take-off. The maximum friction coefficient is $\mu_{r,\max} = 0.7$. First evaluate the deceleration Γ for a velocity of $0.7V_{TD}$, and use that value to evaluate the length needed to decelerate, L_{brake}. Account for the decrease of μ_r with velocity.

Calculate the total runway length needed to land:

$$L_r = L_{\text{descent}} + L_{1/2\text{ flare}} + L_{\text{brake}} \tag{17}$$

and compare with the approximation given by the magic line.

Solutions

For landing with configuration #4 (full deployment):

$$L_r = L_{\text{descent}} + L_{1/2\text{ flare}} + L_{\text{brake}} \tag{18}$$

Descent. Formulation for the distance required is the following:

$$L_{\text{descent}} = \frac{h_s}{\tan|\gamma|} = \frac{15.2}{\tan(3^\circ)} \approx 290 \text{ m} \tag{19}$$

with $\gamma = -3^\circ$ (ILS slope).

Flare the (half) flare is

$$L_{1/2\text{ flare}} = \frac{1}{n_f - 1}\frac{V_{\text{app}}^2}{2g}\tan|\gamma| = \frac{1}{0.1}\frac{(1.3V_S)^2}{2g} \times \tan(3^\circ) \tag{20}$$

Braking.

- Deceleration: $\Gamma = -\mu_r g - \frac{\frac{1}{2}\rho S V^2}{m}(C_x - \mu_r C_z)$ (thrust = 0).
- Friction decreases with velocity: $\mu_r = \frac{\mu_{r,\max}}{1+V/W_r}$ with $\mu_{r,\max} = 0.9$, $W_r = 100$ m/s (dry concrete).
- During braking: landing gear + spoilers: $\Delta C_{x_0} = 0.0175 + 0.06 = 0.0775$, $\Delta C_{z,\text{SP}} = -0.15$.

$$L_{\text{brake}} = \frac{V_{\text{TD}}^2}{2|\Gamma_{0.7V_{\text{TD}}}|} \tag{21}$$

Results. For configuration #4: $V_S = 48.6$ m/s, $V_{\text{app}} = 63.1$ m/s, $V_{\text{TD}} = 55.9$ m/s.

Phase	Length (m)
L_{descent}	290
$L_{1/2\text{ flare}}$	107
L_{brake}	363
L_r	760

The magic line typically gives $L_r \approx 0.3V_S^2$ to $0.4V_S^2$ (in m), yielding $\approx$ 710–945 m for this case. Our result of 760 m is consistent with that range.